

Jianfeng Zhang

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EDUCATION

- **Wuhan University**

Wuhan, China

Bachelor of Science in Applied Mathematics; GPA: 3.64; Ranking: 3/42

Sep. 2014 – Jun. 2018

RESEARCH EXPERIENCE

- **Image Segmentation Algorithm Based on Alternating Descent Solver**

Wuhan University, China

Research Assistant

Mar. 2018 - Present

- Build a Mumford-Shah-like model framework for interactive image segmentation.
- Apply geodesic distance in LAB space to build unary term and utilize L1, L2 total variation, and fast-bilateral-solver-like regularization as pairwise term.
- Develop an approach to optimize unary and pairwise terms alternately to perform image segmentation. Our algorithm can reach 0.8 mean IoU in graphcut dataset.

- **Long Short-Term Memory (LSTM) for Water Table Depth Prediction**

Wuhan University, China

Research Assistant

Apr. 2017 - Jan. 2018

- Developed a LSTM-based model to forecast water stable depth in Hetao Irrigation District. The final best model achieved 0.93 R^2 score on prediction tasks.
- Built a GUI to construct a decision system based on this model, to aid engineers and decision-makers to plan and manage groundwater resources in Hetao Irrigation District.

PUBLICATIONS

Jianfeng Zhang, Yan Zhu, Xiaoping Zhang, Ming Ye and Jinzhong Yang. “*Developing a Long Short-Term Memory (LSTM) based model for predicting water table depth in agricultural areas*”. In: *Journal of Hydrology* (2018). [[Paper](#)] [[Code](#)].

WORK EXPERIENCE

- **Farsee2 Inc.**

Wuhan, China

Research Intern with Dr. [Xiaoping Zhang](#)

Nov. 2017 - Present

- Build a video object segmentation framework based on CNN and user interaction, this framework is currently embedded in our products.
- Review current semantic and instance segmentation models and build a model based on deformable convolutional network and ConvCRF for instance segmentation.

PROJECTS

- **ICPR MTWI 2018 Challenge on Tianchi**

Mar. 2018 - May. 2018

- Built a faster-rcnn like model for end to end network image text detection and recognition. We Ranked 13/1488 on this [challenge](#).

- **Machine Learning Seminar**

Oct. 2017 - Present

- Organizing weekly machine learning seminar in Wuhan University, including professors and students.
- Gave talks about convex optimization, deep learning and their applications.

TEACHING EXPERIENCE

- **C Programming Language**

Spring 2017

- **Data Structures and Algorithms in Python**

Fall 2017

PROGRAMMING SKILLS

- **Languages:** Python, MATLAB, C/C++, R

- **Deep Learning Tools:** PyTorch, Tensorflow, Keras